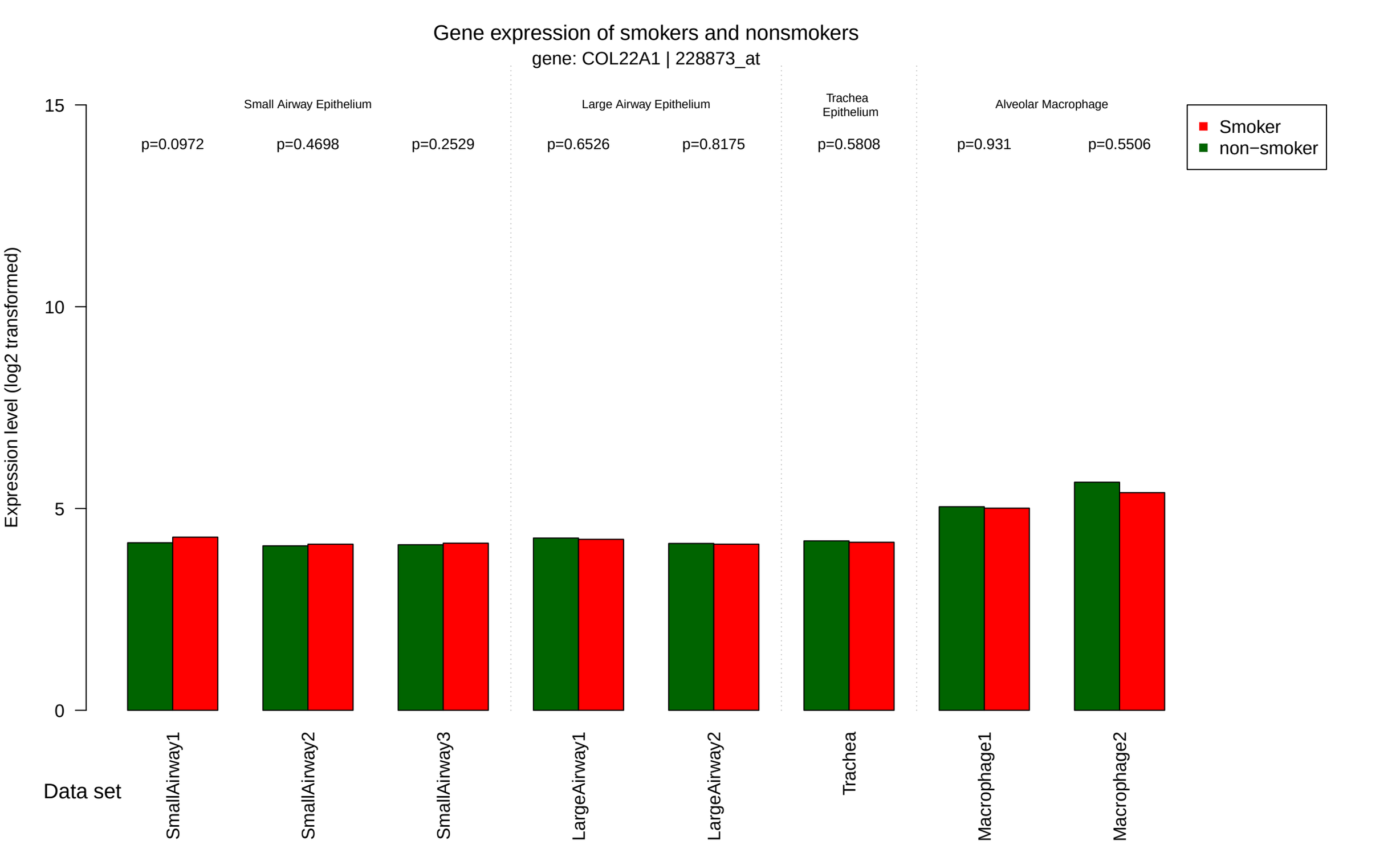
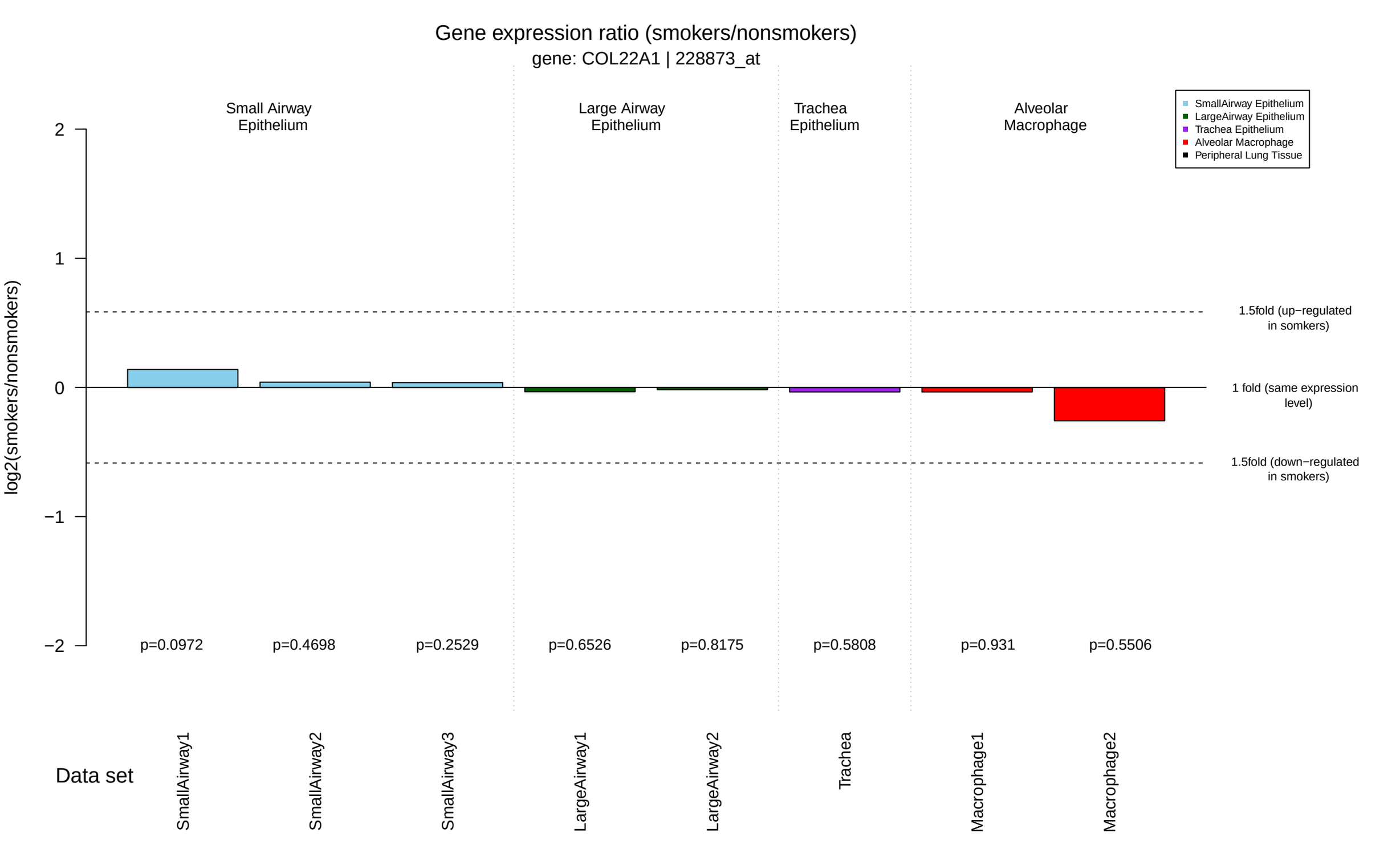
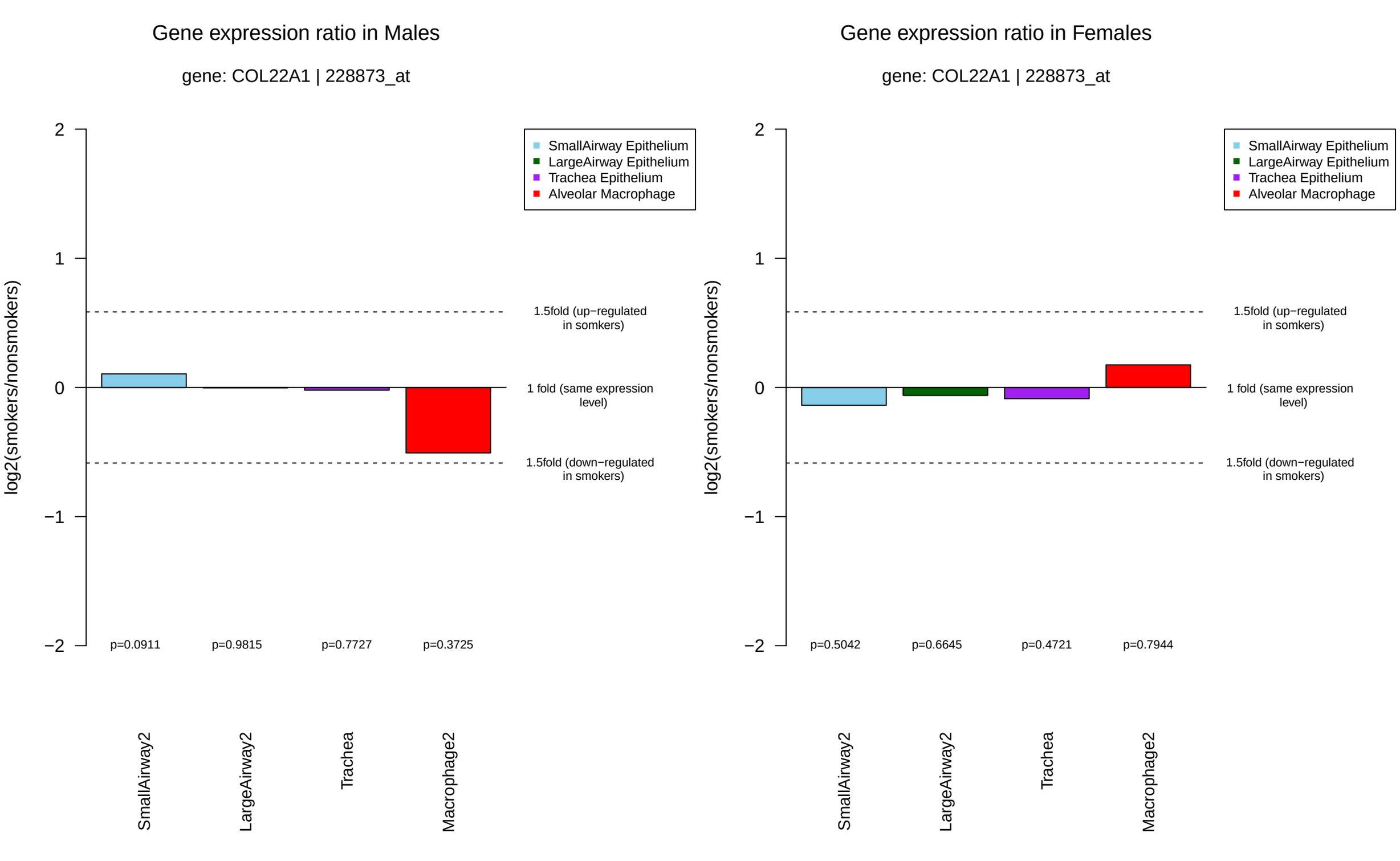
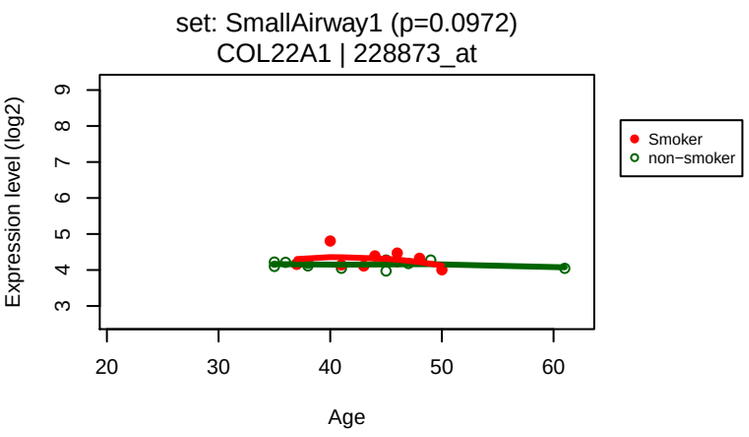
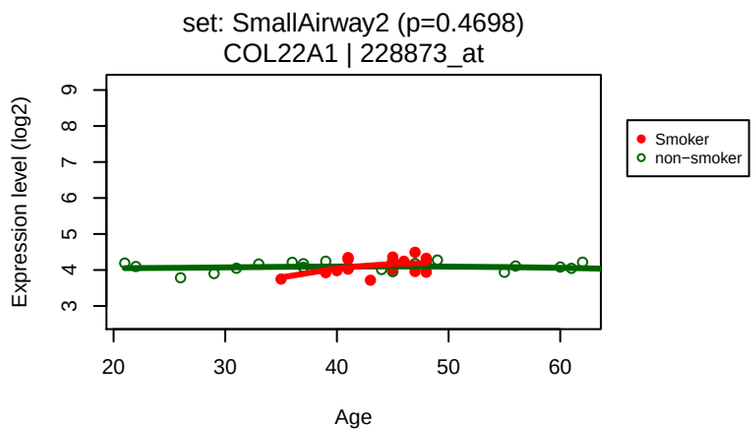
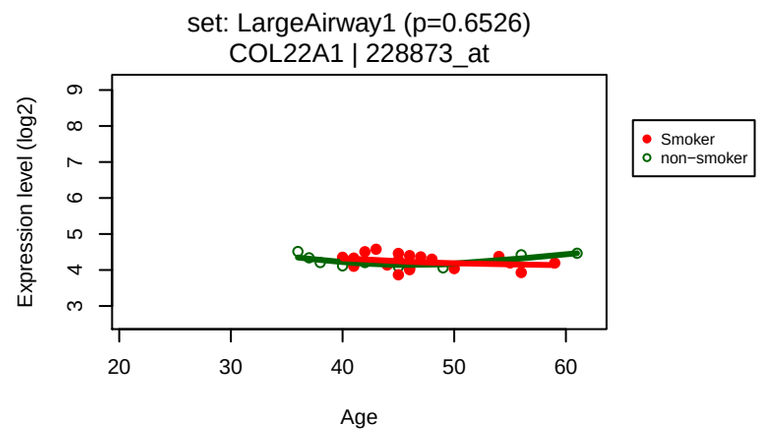

Gene expression of smokers and nonsmokers
gene: COL22A1 | 228873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0972
p=0.4698
p=0.2529
p=0.6526
p=0.8175
p=0.5808
p=0.931
p=0.5506
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

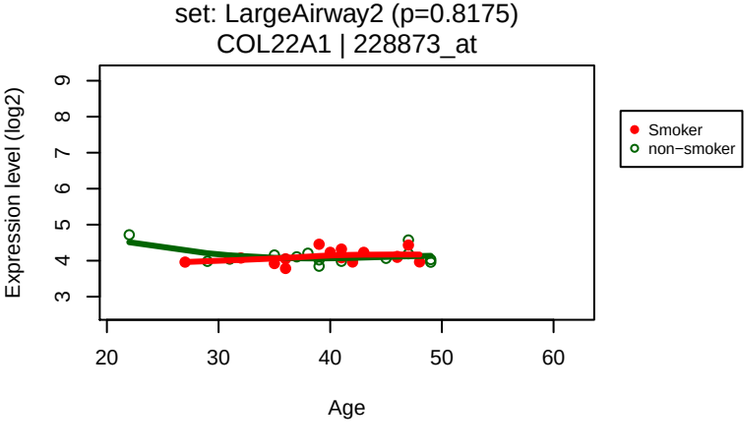

Gene expression ratio (smokers/nonsmokers)
gene: COL22A1 | 228873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.4698
p=0.2529
p=0.6526
p=0.8175
p=0.5808
p=0.931
p=0.5506
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

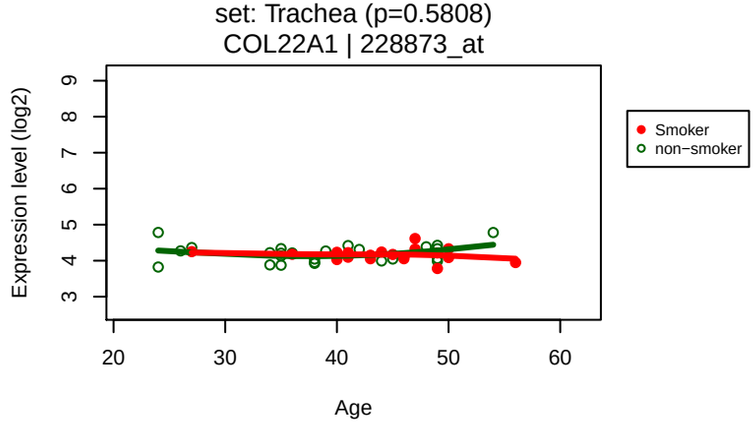

Gene expression ratio in Males
gene: COL22A1 | 228873_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0911
p=0.9815
p=0.7727
p=0.3725
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL22A1 | 228873_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5042
p=0.6645
p=0.4721
p=0.7944
SmallAirway2
LargeAirway2
Trachea
Macrophage2

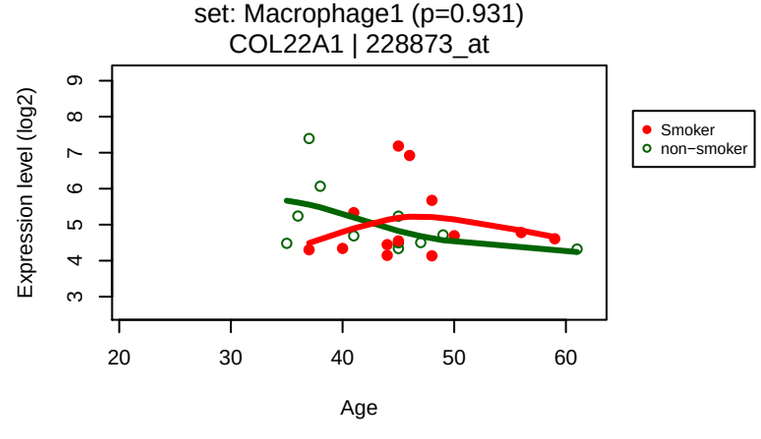

set: SmallAirway1 (p=0.0972)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker

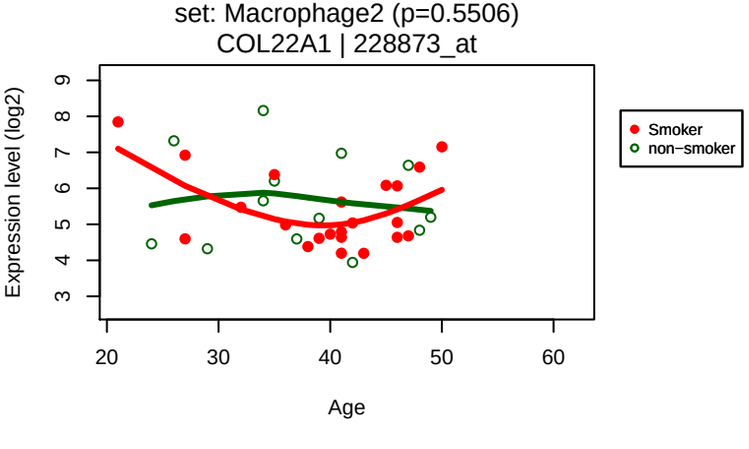

set: SmallAirway2 (p=0.4698)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker

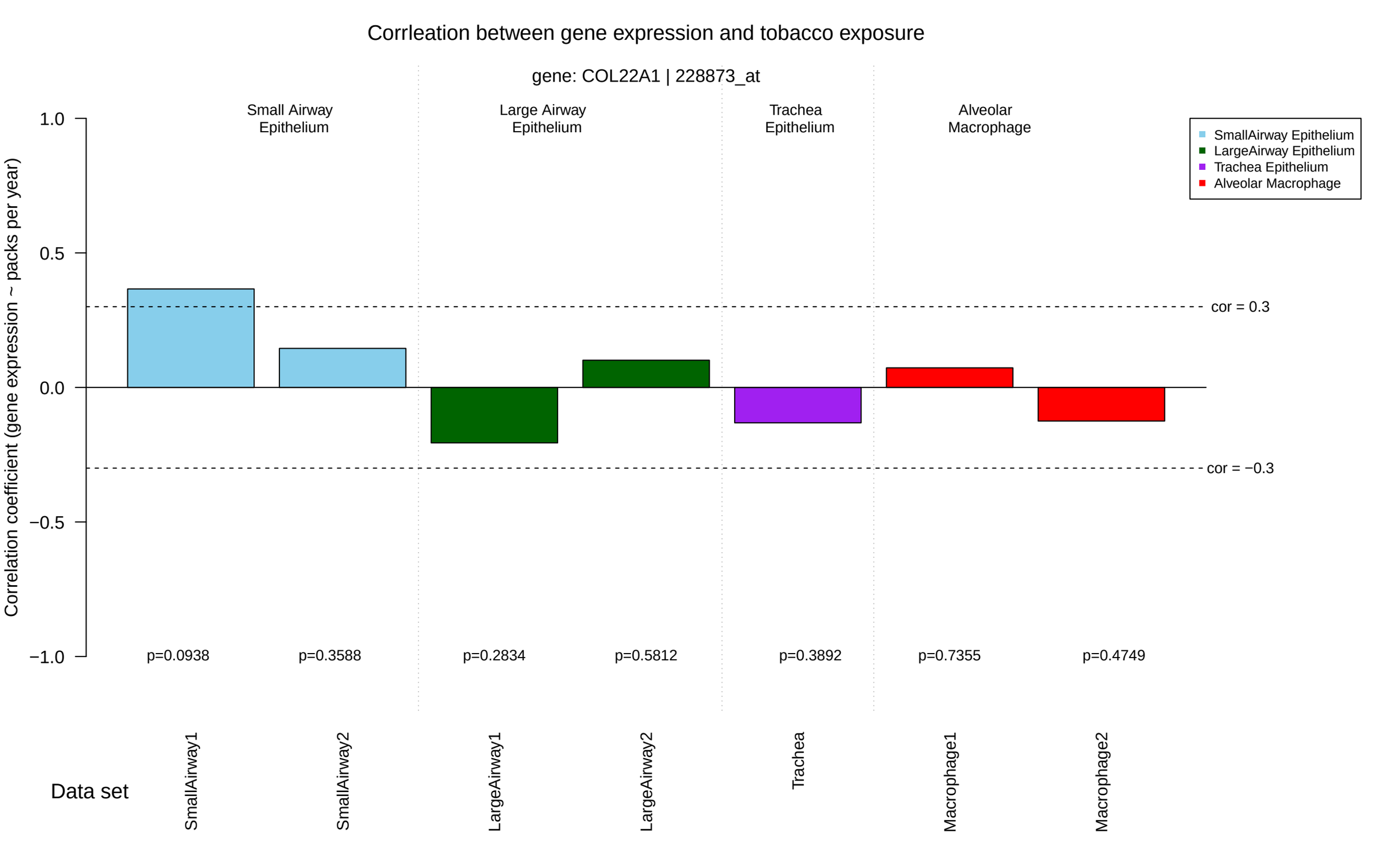

set: LargeAirway1 (p=0.6526)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8175)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5808)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.931)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5506)
COL22A1 | 228873_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL22A1 | 228873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0938
p=0.3588
p=0.2834
p=0.5812
p=0.3892
p=0.7355
p=0.4749
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2